Do Older Directors Make Bigger Hits? Box Office for Films Based on Video Games, 1993–2025

Course example · ECON 3916 Midterm Project · Dr. Richeng Piao

Repository: github.com/RyanPiao/econ3916-midterm-example **Notebook:** colab.research.google.com/github/RyanPiao/econ3916-midterm-example/blob/main/midterm-example-movies.ipynb

Question	Among English-language films based on video games, 1993–2025, do films by older directors earn more at the worldwide box office, holding the release year and the format (live-action or animated) constant?
Headline finding	A director one year older goes with a 2.4% higher worldwide gross (95% CI −3.0% to +8.0%, N = 51), so the data cannot rule out no association. Holding the budget fixed changes the sign of the point estimate, but neither estimate is distinguishable from zero. This is a conditional association, not the effect of experience.
Data	One row is one film (51 in the sample, from 68 listed). Budgets and grosses: Wikipedia's "List of films based on video games", each figure citing Box Office Mojo or The Numbers, downloaded 2026-11-03. Directors' birth dates: Wikidata. All dollars in 2024 prices with the CPI-U from FRED.
Method	OLS of the log of worldwide gross on the director's age at release, the release year and an animated indicator, with robust (HC1) standard errors.

1. The Question and the Data

When a studio adapts a video game, the property is fixed before anyone is hired: the characters, the story and the fan base come with the license. The director is one of the few large choices left, and experience is expensive. An established director costs more and is usually trusted with a bigger budget. If films by older, more experienced directors earned more, a studio would have a reason to pay for that experience. If they did not, the money might be better spent elsewhere. The question matters to the studio executive making that hire, and to anyone who has wondered why adaptations of the same kind of game have earned anywhere from almost nothing to over a billion dollars.

My question is whether, among English-language films based on video games released from 1993 to 2025, films by older directors earn more at the worldwide box office, holding the release year and the format constant. One row is one film. The outcome is the film's worldwide gross in millions of 2024 dollars, which I model in logs. The main explanatory variable is the director's age at release in years: the release year minus the director's birth year. The two controls are the release year and whether the film is animated.

The data are a product of several measurement decisions. The list of films comes from Wikipedia's "List of films based on video games", which volunteer editors keep up to date. They did not measure anything themselves: every budget and gross in the table cites Box Office Mojo or The Numbers, two trade sites that collect the figures studios and distributors report to the industry press. A gross is ticket sales, not profit, and the worldwide figures for older films and small markets are the least complete. A budget is

what the studio announced. It is often rounded or given as a range, it leaves out marketing, and studios have reasons to round it in either direction. The directors' birth dates come from Wikidata, another volunteer database, which I matched to the films by the director's name. The price index is the BLS consumer price index for all urban consumers (CPIAUCSL), taken from FRED, which I used to put every dollar in 2024 prices: a 1993 dollar and a 2023 dollar do not buy the same thing.

Building the sample took four steps, and each one changed the data. The page lists 68 films in two tables, 55 live-action and 13 animated. The title is not a key: Mortal Kombat, Resident Evil and Street Fighter were each remade, so I identify a film by its title and release year, which has no duplicates. Budgets and grosses arrive as text ("$42–48 million[2]"); I took the midpoint of a range and removed commas and footnote marks. One budget, for Moshi Monsters, is given in pounds, and I set it to missing rather than mix currencies. Finally, merging the films with the birth dates on the director's name first matched only 58 of 68 films. Reading the list of unmatched names showed that films with two directors came out of the web table with the names glued together, so I matched on the first director listed. After that, 6 films had no match and 2 more had a match with no birth date on record.

Three kinds of data are missing, for different reasons. Seven budgets are missing: 6 of the 13 animated films and 1 of the 55 live-action films, most of the animated ones computer-animated Resident Evil spin-offs. The gaps line up with the format, which I observe, so missing at random (MAR) given the format is the label I can defend. The main model does not use the budget, but I drop these films too, so that all three columns of Table A1 use the same films. Most of them would leave anyway for a missing birth date, so the rule costs 3 films, all of them animated; that is one reason only 5 animated films remain and the animated coefficient in Section 3 is so imprecise. Section 4 fits the main model without this rule as a check. 8 directors' ages are missing, and those directors made the smallest films: the median gross is $152.0 million where the birth date is known and $2.5 million where it is not. So these gaps are not missing completely at random. They depend on the gross, which I observe, so MAR given the gross is the defensible label. They may also depend on the director's age itself, since a director with no recorded birth date is often early in a career, which would make them missing not at random (MNAR), but these data cannot test that. What it does to my answer: the sample leans toward bigger releases, and the estimate describes films whose directors are well enough known to be recorded. Finally, the 6 films released in 2026 are still in theatres, so their grosses are not finished; a rule I wrote down before fitting any model drops them. Table 1 shows each step from 68 films to 51. It counts each film once, at the first rule it fails, so only 3 of the 8 films without a birth date appear on its last line. Table 2 summarizes the sample, 5 of whose films are animated.

Table 1. From 68 listed films to the sample

Step	Films
Films listed	68
Released in 2026 (run not finished)	−6
No worldwide gross	−1
No budget in US dollars	−7
No birth date for the director	−3

Films in the sample	51

Steps are applied in order, so a film missing two things is counted once, at the first step it fails.

Table 2. Summary statistics, 51 films, 1993–2025

Variable (units)	Mean	SD	Min	Max	N
Worldwide gross ($m, 2024)	232.1	243.7	0.2	1,401.0	51
Budget ($m, 2024)	85.7	58.9	7.5	251.7	51
Director's age at release (years)	44.7	7.7	30.0	68.0	51
Release year	2011.6	8.9	1993	2025	51
Animated (share of films)	0.10	0.30	0	1	51

Gross and budget in millions of 2024 dollars (CPI-U). Sources: Wikipedia (citing Box Office Mojo and The Numbers), Wikidata, FRED. Downloaded 2026-11-03.

2. Descriptive Evidence

Before fitting a model I looked at the outcome. Gross is heavily skewed: the median film earned $179.6 million, the mean is $232.1 million, and one film, The Super Mario Bros. Movie, earned $1,401.0 million. The skewness is 2.35. On that scale a regression would be driven by a handful of hits, and a dollar difference means something different for a $10 million film than for a $1 billion one. My question is about proportional differences, so I use the log of gross. The right panel of Figure 1 shows that the log fixes the right tail and exposes a left one: 3 films (Postal, Far Cry, Werewolves Within) earned under $5 million because they barely opened in theatres. I keep them in the main sample, since they are real releases, and check in Section 4 whether they drive the result.

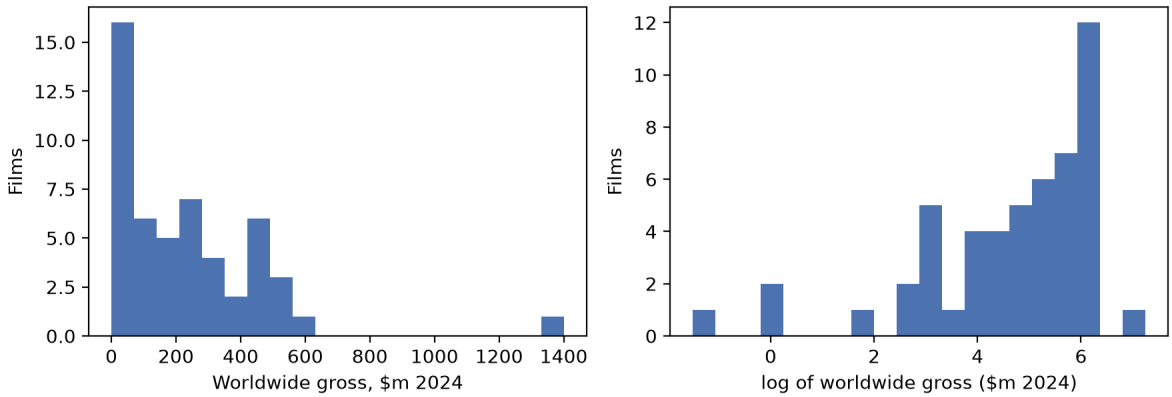


Figure 1. Gross is skewed right: one film earned $1,401.0 million while the median earned $179.6 million. On the log scale the skew reverses, because 3 films barely opened. 51 films, 2024 prices. Source: Wikipedia list (Box Office Mojo, The Numbers); CPI-U from FRED.

Figure 2 plots each film's gross against its director's age at release, with gross on a log scale labelled in dollars so that the axis stays readable. Directors range from 30 to 68 years old. There is no visible slope. The biggest hits and the smallest grosses both come mostly from directors in their forties, which is also where most of the films are. If age mattered a great deal, the cloud would tilt; it does not, and the regression below puts a number and an interval on that impression.

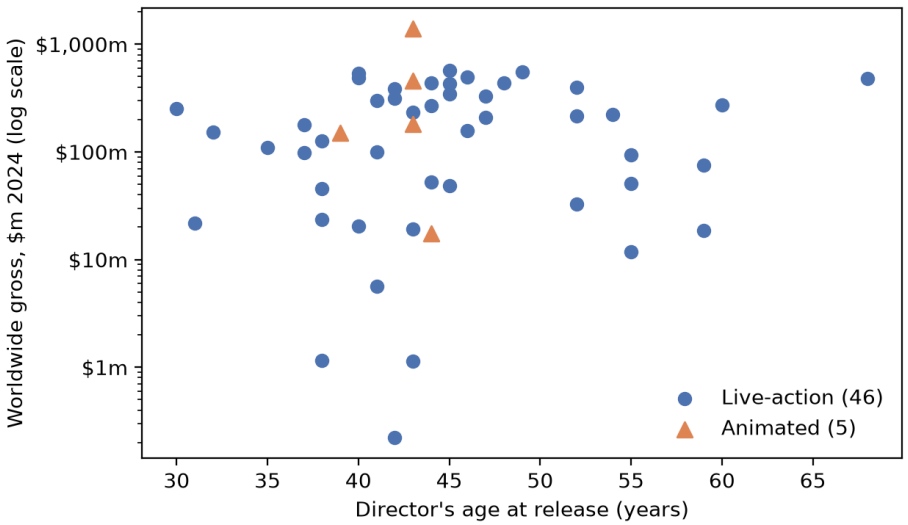


Figure 2. Across directors aged 30 to 68 at release, grosses show no visible trend with age, for live-action or animated films (triangles, 5 films). Each point is one film; the vertical axis is on a log scale. Source: as Figure 1; ages from Wikidata.

3. The Model

I fit the model below by OLS, with heteroskedasticity-robust (HC1) standard errors:

$$log(Gross_i) = \beta_0 + \beta_1 Age_i + \beta_2 Year_i + \beta_3 Animated_i + u_i$$

The full results are in Table A1 in the appendix; column 2 is the main model. I convert log coefficients to percent changes with 100·(e^β − 1) and transform the confidence limits the same way. The coefficient on age is 0.024 with a standard error of 0.027. Because the outcome is in logs, holding the release year and the format fixed, a director one year older goes with a worldwide gross about 2.4% higher, with a 95% confidence interval from −3.0% to +8.0%. The interval includes zero, so this sample cannot rule out no association. It also reaches a 8.0% rise per year, so it cannot rule out a sizeable positive one either. With 51 films the data cannot tell these apart. Without any controls (column 1) the estimate is 2.7% per year, so adding the controls barely moves it.

The second coefficient I interpret is the animated indicator: 0.683 (standard error 0.717). An animated film grosses about 98% more than a live-action film released in the same year by a director of the same age. The 95% interval runs from −51% to +707%, which is very wide, because only 5 of the 51 films are animated. I would not put weight on the size of this number, only on the fact that the model needs to separate the two formats. The release year coefficient is 0.033: each later year goes with 3.3% more gross in real terms (95% CI −1.4% to +8.3%). The R^2 is 0.06: age, year and format explain little of why one film earns more than another. That does not undermine the question, which is about one coefficient, but it says most of what drives box office is outside this model.

Each control has a role:

- **Release year** is there to block a confounding path. Box office for adaptations changed over three decades as the international market grew and as studios moved to franchises, and the ages of the directors studios hire may have changed over the same years. Without it, a trend in both would look

like an effect of age. It would be a bad control only if a director's age could change a film's release year, which it cannot.

- **Animated** also blocks a confounding path. Animated and live-action films are made by different groups of directors and earn on different scales. The format is set by the studio before the director is hired, so age cannot cause it; it would be a bad control only if it could.
- **Budget** is left out on purpose. Older directors get bigger budgets: holding year and format fixed, the budget rises 3.3% for each year of the director's age, and a bigger budget raises the gross. That puts the budget on the path from age to gross: it is a mediator of age's total effect, and for my question a bad control. Column 3 of Table A1 adds it anyway. The point estimate on age changes sign, from +2.4% to −2.7% per year (95% CI −6.8% to +1.5%), and the R^2 rises to 0.45, but neither age estimate is distinguishable from zero, so the sign change is not a finding in itself. Column 3 answers a different question, whether older directors earn more with the same budget (the direct path only). My headline is column 2, the total association.

4. Diagnostics

I checked the main model's assumptions with the tests from Topic 14. First, the residuals. Figure A1 in the appendix plots them against the fitted values and against the director's age. Neither plot shows a fan shape, which would suggest unequal variance, or a curve, which would suggest that a straight line in age is the wrong shape. The one pattern is 3 points far below the rest at low fitted values: the films that barely opened. The residual standard deviation is 1.77 log points, which matches the small R^2: grosses vary a great deal around the model's predictions.

Second, constant variance. The Breusch–Pagan test gives p = 0.88: no evidence of unequal variance. With 51 films the test has little power, so this is weak reassurance, and the residual plots carry more weight. The choice of standard errors is visible in the age coefficient: with classical standard errors its standard error is 0.033 and its 95% interval runs from −4.2% to +9.5%; with robust (HC1) standard errors it is 0.027 and −3.0% to +8.0%. The interval narrows a little and still includes zero, so the conclusion does not depend on the choice. I report HC1 in the main table because it does not require constant variance.

Third, collinearity. The variance inflation factors are 1.03 for age, 1.04 for the release year and 1.03 for the animated indicator. Values this close to 1 are what three nearly unrelated regressors give: in these 51 films, older directors are not concentrated in particular years or in one format. So the wide interval on age does not come from overlap among the regressors. It comes from the small sample and the spread of grosses around the model (a residual standard deviation of 1.77 log points).

Fourth, influence. Using Cook's distance with the common cutoff of 4/n = 0.078, 4 films stand out. The three largest are Ratchet & Clank (0.156), Werewolves Within (0.134) and Postal (0.082); the fourth, The Super Mario Bros. Movie (0.079), sits just over the cutoff. Werewolves Within and Postal opened in very few theatres and earned under $5 million. Dropping them because they are inconvenient would change the question, since they are real releases, so I keep them in the main model and report the estimate without the 3 films under $5 million (Postal, Far Cry, Werewolves Within): the age coefficient goes from 2.4% to 0.1% per year (standard error 2.4 percentage points, N = 48). The estimate moves toward zero, and the reading of it, no clear association, does not change.

Last, the sample rule. Fit on all 54 films released by 2025 with a gross and a director's age, including those with no dollar budget, the age coefficient is −0.9% per year (95% CI −7.7% to +6.4%). The point estimate moves, but the reading does not: no clear association.

5. Threats and Conclusion

For the age coefficient to be causal, nothing else that moves a film's gross could vary with its director's age once the year and the format are fixed, and in this list franchises break that. Resident Evil, Mortal Kombat and Sonic each have several films here, and studios hand the next film in a successful series to a director they already trust, often an experienced one. Those films would earn more whoever directed them, which pushes the age coefficient up. Pulling the other way, a studio may give a first-time director a cheap adaptation of a cult game, which earns little whatever the director's age. My data have no franchise or sequel variable, so neither path is blocked.

Two other threats come from the data. Birth dates are missing for the directors of the smallest films, so the 51 films lean toward bigger releases, and the result describes films whose directors are on record. And the grosses and budgets are studio announcements collected by trade sites: a budget given as "$42–48 million" is entered here as its midpoint.

So the estimate is a conditional association, not a causal effect. Among these 51 films, holding the release year and the format fixed, a director one year older goes with a 2.4% higher worldwide gross, with a 95% interval from −3.0% to +8.0%. A reader should take one thing from it: in these films there is no clear relationship between a director's age and box office, and associations larger than about 8% per year are unlikely. With another month I would add a franchise flag and a sequel flag for each film from the same Wikipedia pages, which would block the path above; have a second person check which films count as "based on a video game", since the list mixes direct adaptations with loose ones; and add the 2026 films once their grosses are final.

References

Federal Reserve Bank of St. Louis. Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (CPIAUCSL), from the U.S. Bureau of Labor Statistics. FRED. Retrieved 2026-11-03.

Wikidata. Date of birth (P569) for each director, retrieved through the Wikidata API on 2026-11-03.

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Appendix

Table A1. Log worldwide gross regressed on the director's age and controls

	(1) Age only	(2) Main	(3) Adds budget
Director's age (years)	0.026	0.024	-0.028
	(0.025)	(0.027)	(0.022)
Release year		0.033	0.049
		(0.024)	(0.021)
Animated (0/1)		0.683	0.039
		(0.717)	(0.453)
log budget			1.571
			(0.294)
Constant	3.448	-62.267	-98.586
	(1.183)	(48.203)	(41.966)
N	51	51	51
R^2	0.01	0.06	0.45

Note: Dependent variable: log of worldwide gross, millions of 2024 dollars. OLS with robust (HC1) standard errors in parentheses. Column 2 is the main model. Column 3 adds the log budget, a mediator (Section 3), for comparison only.

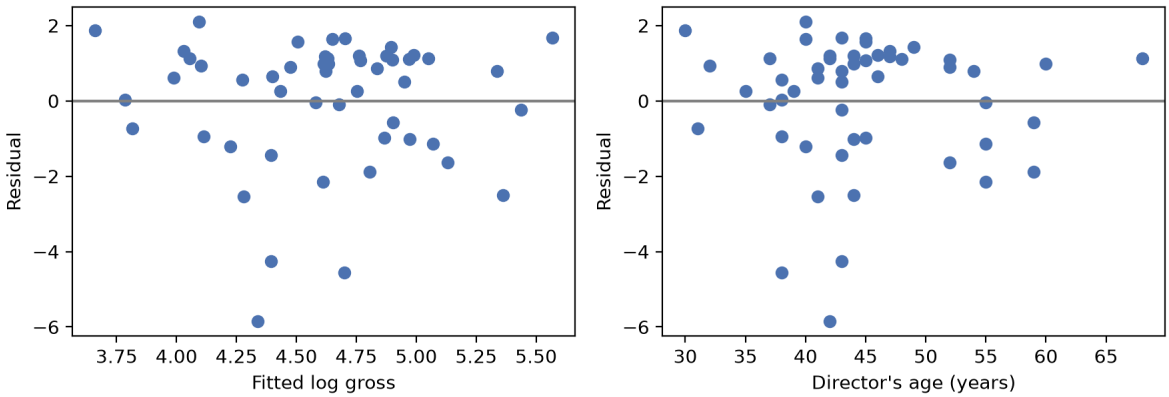


Figure A1. The main model's residuals show no fan and no curve against the fitted values (left) or the director's age (right); the points far below are the films that barely opened.

Reproducing the numbers. Every number in this report is printed by the last code cell of the notebook in the repository, "Numbers my report quotes". The README gives the download step and the order to run the cells.